

LAB 2: DOMAIN SELECTION AND PROTOTYPE DEVELOPMENT

Domain: Agentic Knowledge & Research Intelligence

1. Domain and Problem Statement

Domain:

Agentic Knowledge & Research Intelligence

Problem Statement:

In today's digital environment, users obtain knowledge from multiple sources such as YouTube videos, PDF documents, research papers, and other digital resources. Finding relevant information, understanding large amounts of content, and connecting information across different sources can be time-consuming and inefficient when performed manually.

Therefore, there is a need for an intelligent system that can organize and retrieve information from multiple knowledge sources and assist users in understanding and researching the collected knowledge efficiently.

2. Target Users

The proposed system is primarily targeted towards:

- **Students** who use lectures, YouTube videos, PDFs, and other learning resources.
- **Researchers** who need to analyze and retrieve information from research papers and other documents.
- **Developers** who work with technical documentation, tutorials, and other technical resources.
- **Educators** who manage and use multiple teaching and learning resources.

The primary target users are students and researchers who need to efficiently retrieve, understand, and analyze information from multiple knowledge sources.

3. Proposed Solution with AI/ML Opportunity

Proposed Solution:

The proposed solution is an Agentic Knowledge & Research Intelligence platform that allows users to provide and manage knowledge from multiple sources such as YouTube videos, PDFs, documents, and other supported knowledge bases.

The system will process the provided knowledge and use an AI agent with Retrieval-Augmented Generation (RAG) to retrieve relevant information and generate responses based on the available knowledge. The agent can assist users in performing research-oriented tasks such as asking questions, summarizing content, comparing information from different sources, and extracting useful insights.

The system will progressively incorporate agentic capabilities such as reasoning, task planning, knowledge-source selection, and tool usage to improve the research process.

AI/ML Opportunities:

- Retrieval-Augmented Generation (RAG): Retrieve relevant information from the user's knowledge base before generating responses.
 - Intelligent Search: Search for information based on semantic meaning rather than only matching keywords.
 - Text Summarization: Generate concise summaries of PDFs, YouTube transcripts, and other documents.
 - Knowledge Extraction: Extract important concepts and information from different knowledge sources.
 - Recommendation: Recommend related knowledge or resources based on the user's research context.
 - AI Assistant / Agent: Provide an intelligent research assistant capable of reasoning over retrieved information and assisting with research tasks.
 - Agentic Reasoning and Planning: Break complex research requests into smaller tasks and determine appropriate actions or knowledge sources.
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Proposed Full Stack Application:

The application will be developed as a full-stack web application that manages users, knowledge sources, documents, research sessions, queries, and AI-generated results.

Proposed Technology Flow:

User → React Frontend → Node.js + Express → MongoDB → AI/ML Agent → Vector Database → RAG → LLM → Response

The application can be progressively developed throughout the semester by implementing the frontend, backend, database, authentication, knowledge-source management, RAG functionality, and agentic AI capabilities.